

Hospital Risk Intelligence API Documentation

1. Health Check

Endpoint

GET /health

Sample Response

```
{
  "status": "healthy",
  "model_version": "1.0.0"
}
```

2. Risk Prediction

Endpoint

POST /predict/risk

Sample Request

```
{
  "age": 45,
  "gender": "F",
  "city": "Bangalore",
  "chronic_flag": 1,
  "department": "Cardiology",
  "visit_type": "OPD",
  "length_of_stay_hours": 12,
  "patient_visit_frequency": 3,
  "avg_los_per_patient": 8,
  "days_since_registration": 365,
  "visit_month": 6,
  "visit_day_of_week": 2
}
```

Sample Response

```
{
  "prediction": "Medium",
  "probabilities": {
    "High": 0.27,
    "Low": 0.29,
    "Medium": 0.43
  },
  "model_version": "1.0.0"
}
```

3. Claim Prediction

Endpoint

POST /predict/claim

Sample Request

```
{
  "billed_amount": 10000,
  "approved_amount": 6000,
  "approval_ratio": 0.6,
  "length_of_stay_hours": 3,
  "department": "Cardiology",
  "visit_type": "OPD",
  "age": 10,
  "gender": "F",
  "chronic_flag": 1,
  "insurance_provider": "MediCareX",
  "patient_visit_frequency": 1,
  "days_since_registration": 100,
  "visit_month": 6
}
```

Sample Response

```
{
  "prediction": "Paid",
  "probabilities": {
    "Paid": 0.96,
    "Pending": 0.03,
```

```
    "Rejected": 0.01  
  },  
  "model_version": "1.0.0"  
}
```